

EDA Summary Report

Project Title

Titanic Data Cleaning & Passenger Analysis

Objective

The objective of this project is to clean the Titanic dataset and analyze passenger survival trends using Exploratory Data Analysis (EDA). The analysis helps in understanding the dataset, handling missing values, detecting outliers, and identifying survival patterns.

Dataset Information

- Dataset Name: Titanic Dataset
- Source: Kaggle
- Number of Records: 891
- Number of Columns: 12 (before removing the Cabin column)

The dataset contains information about passengers, including age, gender, passenger class, fare, and survival status.

Data Cleaning

- Checked for missing values.
- Filled missing values in the **Age** column using the median.
- Filled missing values in the **Embarked** column using the mode.
- Removed the **Cabin** column because it contained a large number of missing values.
- Checked for duplicate records and found no duplicate entries.

Outlier Detection

Outliers in the **Age** column were detected using the IQR method.

- Lower Limit = **2.5**
- Upper Limit = **54.5**

A total of **66 outliers** were identified and treated using the capping method.

Data Analysis

The dataset was analyzed using:

- Filtering
- Sorting
- GroupBy
- Survival statistics

Visualizations

The following visualizations were created:

- Bar Chart
- Line Chart
- Histogram
- Scatter Plot

These visualizations helped in understanding passenger distribution and survival trends.

Conclusion

This project improved my understanding of data cleaning, missing value handling, outlier detection, filtering, grouping, and visualization. It also demonstrated how EDA can be used to identify meaningful patterns in a real-world dataset.